CST380 - Mobile & Ubiquitous Computing

CST380 (Mobile and Ubiquitous Computing) Syllabus

Course Information

Course Details

- **Course Title:** Mobile and Ubiquitous Computing
- **Course Number:** CST380

Course Description

Mobile devices are largely ubiquitous in modern society, ranging from cell phones to tablets and VR headsets. Learning how to develop applications for these devices, and learning how to make these applications as functional as possible is an increasingly important skill. In this course, we learn how to develop applications for the iOS platform, as well as how to make design considerations for these platforms. To learn how to develop these applications, students will work through labs and projects to develop the core programming skills for the swift programming language and the iOS platform. To learn how to consider the trade-offs for mobile development, students will learn how to consider the pros and cons of various development choices with regard to mobile devices.

Course Objectives

At the end of this class, you should be able to: - Develop iOS applications using the Swift Programming Language - Understand different UI modalities and their best applications - Be able to research specific topics and apply them to your own projects - Be able to articulate and compare different mobile-oriented design challenges and choices

Topics Covered

The major topics covered in class are: - The swift programming language - The XCode development environment - Mobile UI modalities - Mobile specific trade-offs

Personnel

- Dr. Sam Ogden (instructor)
  - E-mail: `sogden@csumb.edu`
  - Web: **`https://csumb.edu/scd/sogden/`**
  - Office: BIT205
  - Office Hours: 2pm-3pm Mondays & 1pm-2pm Wednesdays, or by appointment

Getting Help: Office hours are available throughout the week - make use of them! You can also use the class communication platform to ask questions, or send me questions via email.

Communicating online I will use email, Canvas, and Slack for online class communication.

1. **email** is to be used when you need to officially communicate something to me.

- e.g. “Can you double check this assignment grade for me?”

2. **Slack** will be used to answer content questions and give quick updates.
   - e.g. “I think I found a bug in this quiz, could you check it out?”
   - We have a slack channel on the CodePath workspace and I’m available on the CS-U-Monterey workspace that you should join.
3. **Canvas** will have information on assignments and due dates
   - In general I don’t read things in canvas messages or comments. For questions or comments please use Slack or email instead.

Materials

- Most all of the assignments are located on CodePath
- Canvas Course Page: CST380 on Canvas
- Syllabus: CST380 Syllabus

Grading

There are three groups of assignments, briefly described in Table 1.

Assignment Kind	Value
Weekly Labs	30%
Weekly Projects	30%
Special Topics (team)	15%
Final Project (team)	20%
Misc	5%

Table 1. Points available per assignment type.

Weekly Labs are small, guided projects that are meant to introduce topics and provide you with the background to complete the weekly projects. They should be turned in on Canvas.

Weekly Projects are projects that you create to showcase the skills learned each week. These are the homework associated with the course. They should be turned in via CodePath and with a screenshot submitted on canvas. Note that on CodePath the scoring for these projects is broken up into “required” and “stretch” goals, with roughly a third of the points being stretch goals. For grading I use a formula that has required points graded at 1 point and stretch points graded at 0.5 points – encouraging, but not requiring, stretch goal completion.

Special Topics are going to be presented part way through the semester. As a small team you will pick a topic of interest in mobile development or iOS programming and present it to the class, along with a small Minimum Working Example. The goal is to get you, and your classmates, interested in something you might use in your final project.

Final Project are group projects we will complete in the second half of the semester, where you develop a project together. This project will be a finished application and will include code, a demonstration, and a presentation highlighting the application you have built.

Misc work will be for tasks that need to be completed, such as signing up for CodePath.

CodePath

CodePath is providing a number of assignments designed to help you learn how to develop applications for iOS. These assignments are submitted through the CodePath portal and are graded by CodePath. Their team grades two days a week (Monday and Wednesday), and so will provide feedback within 24hours of these days. They will also provide feedback and be available for discussion on the CodePath slack workspace.

While they do accept late work and will provide the grade out of 100%, any late work will have its score adjusted by me in canvas per late policy.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below, with exceptions made based on the general grading requirements guidance. Note that $[90, 93)$ means "at least 90 but less than 93" (e.g. $90 \leq score < 93$) and that all grades will be rounded to the nearest whole number (e.g. a 92.1 will be rounded to a 92, but a 92.5 will be rounded to a 93). Additionally, note that there will be no curve.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Course Policies

Attendance

Attending lecture is strongly correlated with academic achievement[1], and thus is strongly encouraged. However, if you are sick, please do not come to class.

Late Policy

Almost all assignments can be submitted later, albeit with a late penalty. This late penalty will be a 10% reduction in maximum points per day, with a maximum reduction of 60%[2].

If you do fall behind in class, please set up a time to meet with me to discuss getting you back on track – my goal is to have you succeed in class and I want to find a way to make that happen.

Academic Honor Code

In addition to the department- and school-wide policies on academic honesty outlined elsewhere on canvas, I wanted to emphasize a few key points about expectations in class. Note that these are different from other classes I teach.

Specifically, I believe that LLMs will be an important tool in our tool kits as computer scientists and that using them to help us understand new topics and modalities will lead to us using them in a way similar to how we already use google and StackOverflow. To this end, I encourage you to interact with ChatGPT and other LLMs to further your knowledge of topics in this class.

Mobile development requires good coding skills and a strong understanding of the trade-offs inherent in mobile-oriented development and interface design. Therefore, while LLMs can help you understand the coding, there are often problems that are beyond its scope.

[1] Attendance and grade are strongly correlated (not just because of the attendance quizzes but they also factor in).

[2] This is done automatically through canvas and it "chops off" the points that you can't get. For example, if you turn in an assignment 1 day late and would have gotten a 95% on it you will get a 90% for it.

University Academic Integrity Policy

For complete academic integrity policies, please refer to the CSUMB Academic Integrity Policy.

University Policies and Resources

Enrollment and Registration

For information about requesting an incomplete or withdrawal from the course, please refer to CSUMB's Enrollment and Registration Policy.

For grade appeals, consult the Grade Appeal Policy.

Disability Services

If you have a disability that may require academic accommodations, please contact the Student Disability Resources office as soon as possible. All discussions will remain confidential. Students who have already received approval for accommodations from SDR should schedule a meeting with the instructor as early in the semester as possible.

Collection of Student Work

Student work may be collected and used for course assessment and improvement purposes. By enrolling in this course, you consent to the collection and analysis of your academic work for educational assessment. All student work will be handled confidentially and in accordance with FERPA regulations.

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.
